Classification with ADALINE Network

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1 Introduction

The ADALINE (Adaptive Linear Element) is a single-layer neural network that uses the Delta Rule for weight updates[cite: 3]. Unlike the Perceptron, the error is calculated based on the continuous activation potential (u), allowing for the minimization of the Mean Squared Error (MSE) through gradient descent[cite: 4]. This report analyzes the behavior of ADALINE using two optimization approaches: Stochastic Gradient (sample-by-sample update) and Batch Gradient (update per epoch)[cite: 5].

2 Dataset 1: Linearly Separable Data

2.1 Error Evolution (MSE and Classification)

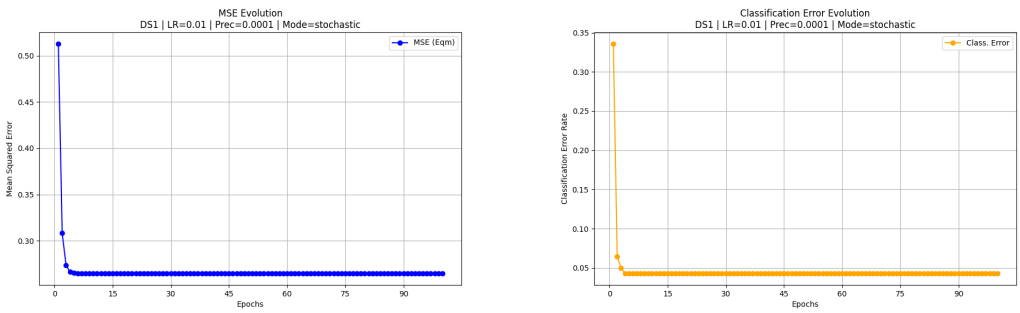

Figure 1: MSE and Classification Error Evolution (Stochastic).

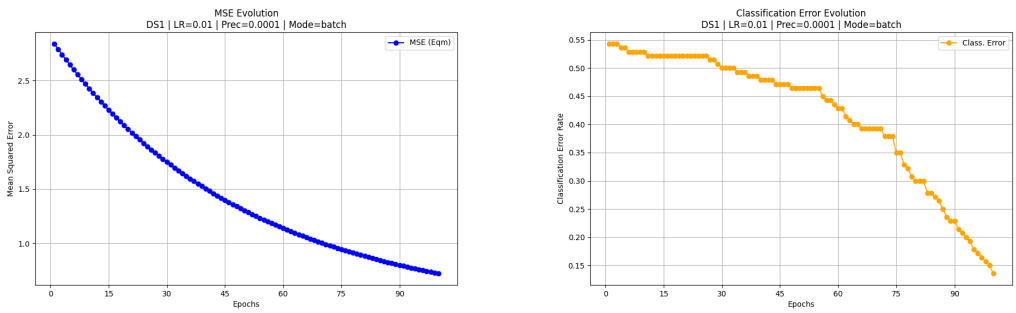

Figure 2: MSE and Classification Error Evolution (Batch).

2.2 Decision Boundaries (Train and Test)

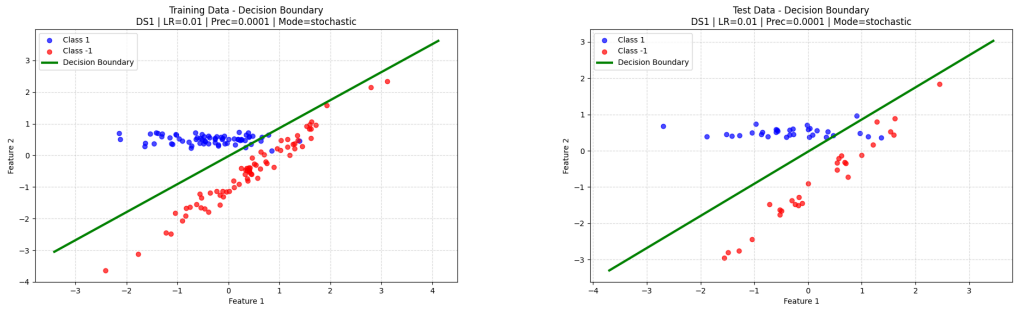

Figure 3: Decision Boundaries in Training and Testing (Stochastic).

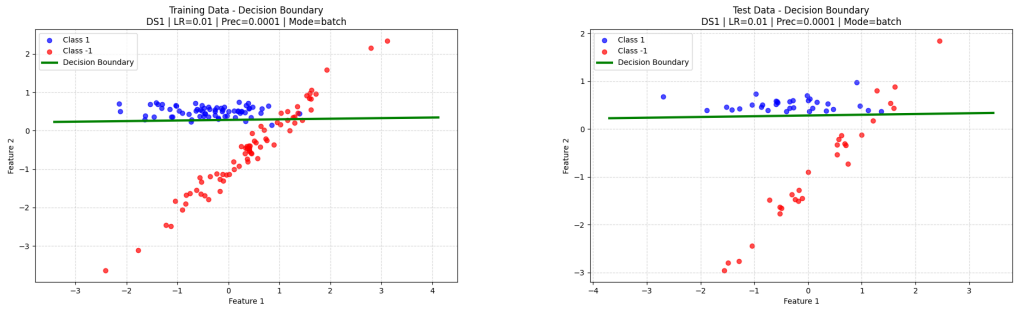

Figure 4: Decision Boundaries in Training and Testing (Batch).

2.3 Discussion of Results

Stochastic Gradient: Showed rapid convergence, adjusting the hyperplane to perfectly separate the classes in a few epochs[cite: 7]. The error descent showed natural oscillations due to the sample-by-sample update[cite: 8].

Batch Gradient: The MSE evolution was notably smoother and more stable[cite: 9]. Since the data is perfectly separable, both methods found an optimal boundary centered between the class margins, generalizing perfectly to the test set[cite: 10].

3 Dataset 2: Non-Linearly Separable Data (Noise)

3.1 Error Evolution (MSE and Classification)

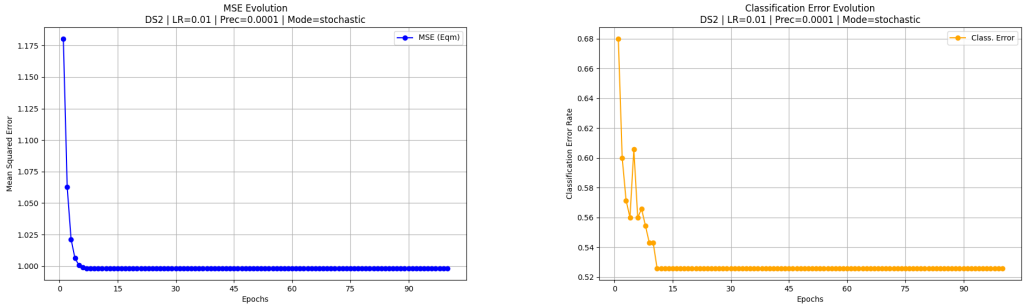

Figure 5: MSE and Classification Error Evolution (Stochastic).

3.2 Decision Boundaries (Train and Test)

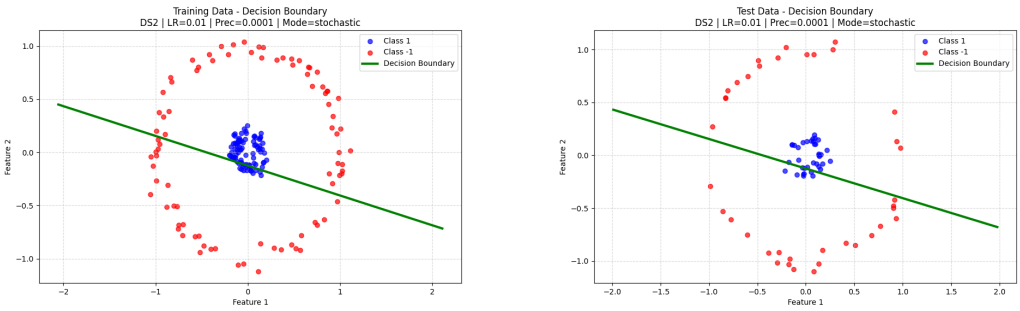

Figure 6: Decision Boundaries in Training and Testing with Noise (Stochastic).

3.3 Discussion of Results

In this dataset, the presence of noise (overlap) prevents the classification error from reaching zero[cite: 12].

Stochastic Gradient: The algorithm continued to attempt adjustments to the noise, which generated noticeable fluctuations in the hyperplane during training[cite: 13]. However, due to the nature of the Least Mean Squares, the model stabilized at the best possible global solution[cite: 14].

Batch Gradient: Proved to be more robust to local noise, as the accumulation of gradients cancels out individual perturbations from overlapping samples[cite: 15]. The final hyperplane accommodated the noise stably, ensuring consistent accuracy in both training and testing[cite: 16].

4 Dataset 3: High Dimensionality and Hyperparameters

4.1 Accuracy in Training and Test Sets

Table 1 presents the final metrics for each combination tested in Dataset 3 (10 input variables)[cite: 17].

Table 1: Accuracies and Convergence Epochs by Configuration (Dataset 3)

Method	**Rate (η)**	**Precision (ϵ)**	**Epochs**	**Train Acc (%)**	**Test Acc (%)**
Stochastic	0.01	0.1	100	93.88	85.71
Stochastic	0.01	0.0001	100	93.88	85.71
Stochastic	0.001	0.1	100	93.88	85.71
Stochastic	0.001	0.0001	100	93.88	85.71
Stochastic	0.0001	0.1	100	87.76	79.37
Stochastic	0.0001	0.0001	100	87.76	79.37
Batch	0.01	0.1	100	75.51	71.43
Batch	0.01	0.0001	100	75.51	71.43
Batch	0.001	0.1	100	51.02	44.44
Batch	0.001	0.0001	100	51.02	44.44
Batch	0.0001	0.1	100	49.66	44.44
Batch	0.0001	0.0001	100	49.66	44.44

4.2 Discussion of Results

In a 10-dimensional scenario, the choice of weight update method and hyperparameter tuning proved decisive for convergence success[cite: 21].

Stochastic Gradient: Proved to be the most effective strategy for this dataset[cite: 22]. With learning rates of $\eta = 0.01$ and $\eta = 0.001$, the model achieved the highest recorded accuracies: 93.88% in training and 85.71% in testing[cite: 1, 23]. The final mean squared error ($MSE \approx 0.32$) was substantially lower than that of the batch method, indicating that sample-by-sample updates allowed ADALINE to navigate the complex error surface of the 10 variables more efficiently[cite: 1, 24]. Even with a lower rate ($\eta = 0.0001$), performance (87.76% training) still outperformed any configuration in Batch mode[cite: 1, 25].

Batch Gradient: Showed inferior performance and greater difficulty in converging within the stipulated 100 epochs[cite: 1, 26]. The best accuracy obtained was only 75.51% in training and 71.43% in testing with $\eta = 0.01$[cite: 1, 27]. By reducing the learning rate to 0.001 or 0.0001, the model suffered severe degradation, with accuracies falling to the 49-51% range and the MSE jumping to values above 4.0[cite: 1, 28]. This suggests that for this higher-dimensional problem, the update per epoch was too slow to adjust weights properly, resulting in a model that failed to learn essential patterns within the available time[cite: 29].

5 Conclusion

This activity demonstrated the structural advantages of the Delta Rule in the ADALINE network[cite: 30]. In noisy data (Dataset 2), ADALINE overcame the classic limitations of the Perceptron by accommodating noise without diverging[cite: 31]. In Dataset 3, the comparative analysis revealed that the Stochastic Gradient method is more effective for high-dimensional problems, achieving significantly higher accuracy and lower MSE than the Batch mode under the same conditions[cite: 1].